**CALL FOR PROPOSALS FOR SEED FUNDING GRANTS IN INFECTIOUS DISEASES AND CANCER**

1. **Proposed project name:** Spatial modeling of enteric and diarrheal disease incidence using wastewater surveillance
2. **List of key project personnel:**

- Dr Raeesa Manjoo-Docrat: Lecturer, Wits, School of Statistics and Actuarial Sciences [PI,ECR]
- Prof Inger Fabris-Rotelli: Professor, University of Pretoria, Department of Statistics, University of Pretoria [Collaborator].
- Prof Pravesh Debba: Council for Scientific and Industrial Research, Smart Places Urban and Regional Dynamics Research Group [Collaborator].
- Mrs Jenny Holloway and Mrs Nontembeko Dudeni-Tlhone NextGen Enterprises and Institutions, Council for Scientific and Industrial Research [Collaborator].
- Dr David Rose: Lecturer, Wits, School of Statistics and Actuarial Sciences [Collaborator].
- Mrs Nada Abdelatif: Biostatistics Research Unit, South African Medical Research Council [Collaborator].
- Ms Renate Thiede and Mrs Rene Stander, Lecturers, Department of Statistics, University of Pretoria [Collaborator, ECR].
- Mr Wouter le Roux and Mrs Lisa Schaefer Water Centre, Smart Places, Council for Scientific and Industrial Research, South Africa: lead in wastewater microbial analysis [Collaborator]
- Mr Jabulani Jele: Smart Places Urban and Regional Dynamics Research Group, Council for Scientific and Industrial Research [Collaborator]
- Dr Neville Sweijd: Alliance for Collaboration on Climate and Earth Systems Sciences, Smart Places, Council for Scientific and Industrial Research, Expertise in Climate Sciences and Climate Services.[Collaborator]
- Dr Isabel Meyer: Principal Researcher, Smart Mobility, Council for Scientific and Industrial Research [Collaborator]
- Dr Sibusisiwe Makhanya: IBM Research, South Africa[Collaborator]
- Dr Chinwe Iwu-Jaja: Centre for Vaccines and Immunology, National Institute for Communicable Diseases (NICD) [Collaborator]
- Mr Nico van Blerk: Ekurhuleni Water Care Company (ERWAT) [Collaborator]

3. **Background and Rationale:**

This project seeks to expand the application of wastewater-based epidemiology (WBE) to address enteric and diarrheal diseases in South Africa. WBE, a powerful tool in monitoring health indices and disease outbreaks, holds significant promise in a country facing an alarming burden of these diseases, with over 1.7 billion annual cases and 525,000 child deaths reported [1]. South African wastewater has revealed the presence of key enteric pathogens, such as Salmonella, Vibrio cholerae, and Hepatitis A [2,3,4], suggesting underreporting in clinical data [5,6]. By utilising WBE, this research aims to provide more accurate, timely data for public health decision-making, enhancing intervention strategies, and contributing to the global understanding of WBE applications.

4. **Study Aim, Hypotheses, Primary (and secondary, where relevant) Objective(s)**

The focus of the research in this proposal for 2024-2025 will be the expansion of wastewater surveillance of SARS-CoV-2 in SA as an early warning system for COVID-19, to enteric and diarrheal diseases. The overarching approach is one of spatial understanding of this data and the prediction of hotspots and disease tracking. The main hypothesis for this project is that molecular markers can be developed and applied for WBE in order to assess the occurrence of enteric pathogens (as an index of disease incidence). This at a range of space and time scales, and modeled to be used as an outbreak early warning system. The objectives are as follows: (i) Development and application of a method for the molecular detection and quantification of Hepatitis A for enteric disease surveillance in wastewater samples. (ii) Perform enteric and diarrheal disease surveillance at three wastewater treatment facilities in Ekurhuleni, South Africa for Salmonella, Vibrio cholerae, and Hepatitis A (facilities already being

sampled and tested for COVID-19). (iii) Statistical methodology development for spatial modeling on multiple spatial nonaligned and disjoint lattice data structures: Prediction of disease location and severity using wastewater collection, taking into account feeder areas for wastewater plants as well as covariate data. (iv) Combine the results of (i) - (iii) to provide mechanisms for early warning detectors through wastewater-based epidemiology, with spatial variability information, with ultimately an effective operational surveillance system.

**5. Study design:**

Building on our successful 2023 wastewater-based epidemiology research, this 2024 proposal aims to expand monitoring to include enteric and diarrheal diseases in SA. Weekly wastewater samples will be collected from three treatment plants in Ekurhuleni Municipality, targeting three priority pathogens: Salmonella, Vibrio cholerae, and Hepatitis A. RNA and DNA extraction will enable real-time polymerase chain reaction (PCR) for quantifying Salmonella spp. and V. cholerae and developing a real-time reverse transcriptase PCR for Hepatitis A. Addressing nonaligned disjoint spatial lattices in the area, the project will devise a spatial autocorrelation measure that considers both distance and polygon shape. Bayesian spatio-temporal models using the INLA package in R will be applied to this complex lattice structure, with remaining challenges including misalignment and space-time interactions. Incorporating clinical data on notifiable medical conditions (Hepatitis A, cholera, and enteric fever) from the National Institute for Communicable Disease (NICD) SA, the project builds on previously established networks for wastewater sample collection. Covariate data such as climate data sourced from StatsSA census data to enhance spatial modeling. This proposal represents a significant advancement in using WBE for monitoring enteric and diarrheal diseases, contributing to better public health decision-making in South Africa.

**6. Describe how findings from this proposal may be pivotal for future proposals for funding to external agencies/foundations/organisations**

The findings from this proposal are pivotal for securing future funding from external agencies, foundations, and organisations because they demonstrate the research's potential to significantly impact disease control and public health. The ability to expand sampling methods, address various diseases, and develop a surveillance dashboard with warning systems showcases the research's practical applications and broad relevance. This compelling evidence of its feasibility and impact increases the likelihood of attracting external funding for further research, enabling the development of critical disease control measures and outbreak response capabilities.

**7. Describe how the proposed project could advance careers of early career researchers**

The proposed project offers valuable opportunities to advance the careers of early-career researchers which is a large make-up of the team as well as the postgraduate students the team supervisors at UP and WITS each year. It provides a platform to gain hands-on experience in multidisciplinary research, fostering skills in data collection, analysis, and communication. Collaboration with seasoned researchers enhances networking and mentorship. Successful project outcomes can lead to publications, conference presentations, and recognition in the field, bolstering their academic credentials. Additionally, involvement in a project with real-world implications, like disease control, enhances the researchers' professional profiles and opens doors for further research opportunities, career progression, and grant funding, ultimately advancing their careers in both academia and public health. This research specifically facilitates the training of analytics and GIS in public health and disease modeling, with more than half the team being females.

**8. Articulate a vision for this proposed pilot project and subsequent larger projects. What is the pathway to life-saving, disease burden reducing outcomes**

The research's significance lies in its potential to drive increased sampling and real-time disease control efforts. By expanding sampling locations and methodology, this work can be applied to address various diseases and substances like endocrine disruptors, antibiotics, and psychoactive drugs, along with water quality assessment and emerging infectious diseases. The outcomes can be integrated into a surveillance dashboard, enabling the public or internal stakeholders to receive warning alerts based on disease sensitivity. This data aids informed decision-making in healthcare settings. Moreover, it supports the development of outbreak response capabilities in high-risk areas, enhancing access to healthcare facilities and logistics infrastructure to reduce fatalities during outbreaks.

### *9.* Provide milestones and relevant timelines:

| Objectives / aims / tasks | | 2024 | | | | | | | | | | | | 2025 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No. | Description | J | F | M | A | M | J | J | A | S | O | N | D | J | F | M | A | M | J | J | A | S | O | N | D |
| 1 | Development of a method for the molecular detection and quantification of hepatitis A for enteric disease | X | X | X | X | X | X | | | | | | | | | | | | | | | | | | |
| 2 | Wastewater data collection | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| 3 | Spatial misalignment methodology development | X | X | X | X | X | X | X | | | | | | | | | | | | | | | | | |
| 4 | Extended methodology development resulting from insight in (3) | | | | | X | X | X | X | X | X | X | X | | | | | | | | | | | | |
| 5 | Dashboard conceptualisation | | | | | | | | | | X | X | X | X | X | X | | | | | | | | | |
| 6 | Dashboard development and testing | | | | | | | | | | | | | X | X | X | X | X | X | | | | | | |
| 7 | Dashboard finalised and training of use | | | | | | | | | | | | | | | | | | | X | X | X | | | |
| 8 | Modeling and analysis to feed into dashboard | | | | | | | | | | | | | | | | | | | X | X | X | X | X | X |

### *10.* Budget

| No. | Expenditure description in full | Quantity | Unit cost (ZAR) | Total (ZAR) |
|---|---|---|---|---|
| 1 | Wastewater testing equipment (once off testing set up) | 1 | R200 000 | R200 000 |
| 2 | Team writing/research workshop: 3 days. Catering and venue per day per person@R1000, travel and accommodation for half the team@R4000+4 nights@R1500 per night: | 1 | R50 000 | R50 000 |
| 3 | Dashboard development/ training/handover | 1 | R10 000 | R10 000 |
| 4 | Local conferences: Local (South Africa) conferences: 4 team members, 4 students: flights @R4000, registration @R6500, accommodation@R8000 (5 nights) = R18500 per year | 4 | R20 000 | R80 000 |
| 5 | International conferences: International Conferences: 2 team members: flights @R20000, registration @R15000, VISA @ R7000 | 2 | R40 000 | R80 000 |
| 6 | Hons/masters students (want students with some computer science background) research assistance funding | 2 | R30 000 | R60 000 |
| | | | **TOTAL** | **R480 0000** |